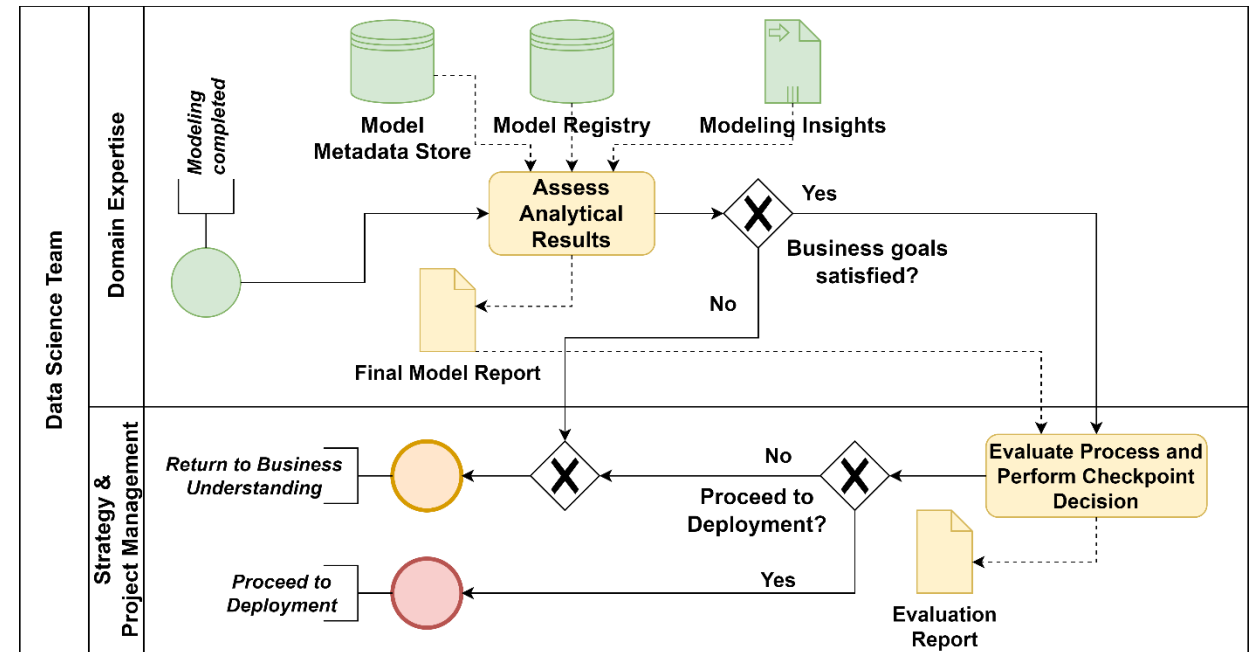


Evaluation

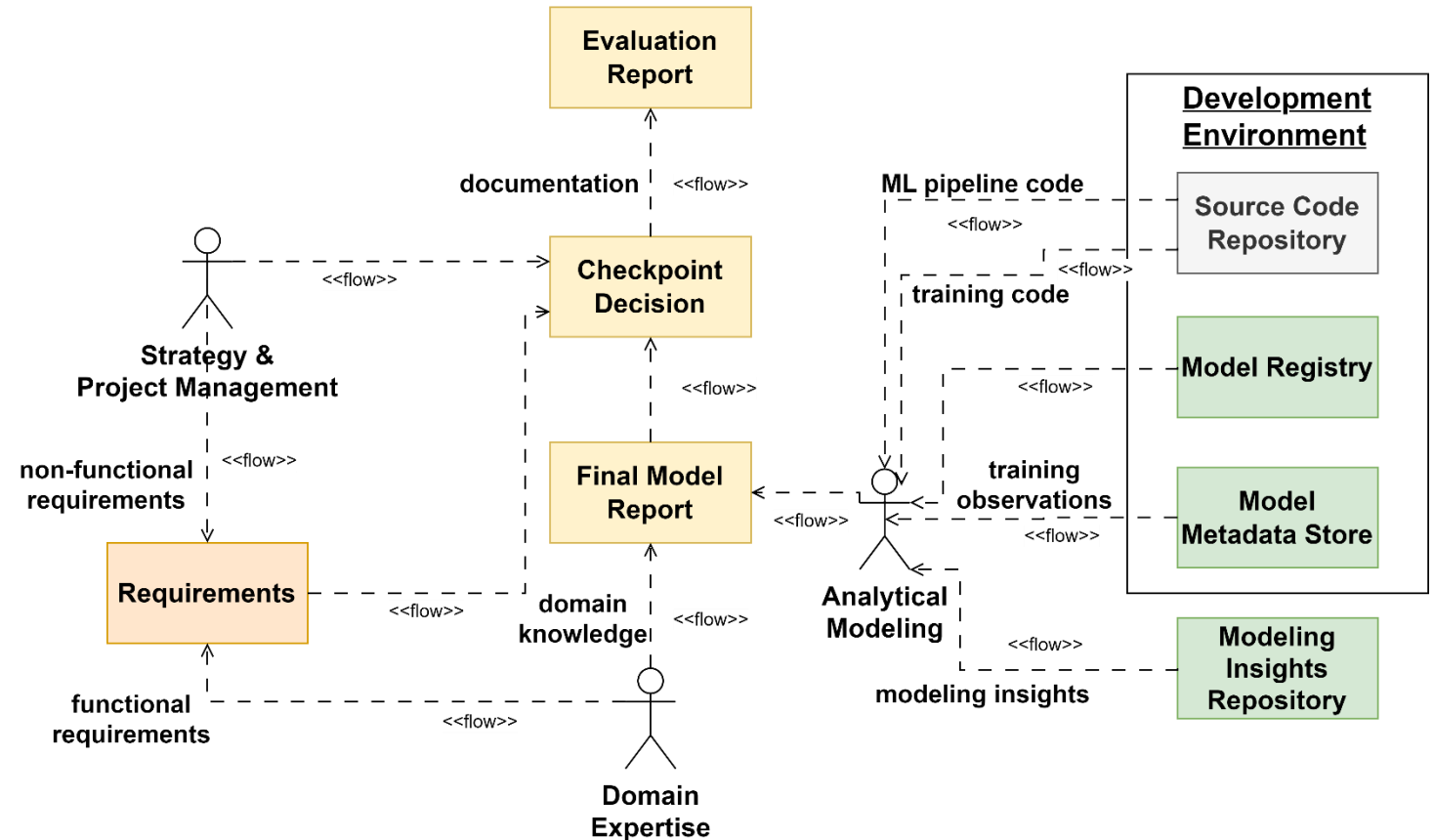
Evaluation – Process Perspective

- Assess the degree to which the developed model(s) (pipeline) meet(s) the business objectives
- Main tasks
 - Assess Analytical Results
 - Evaluate Process and Perform Checkpoint Decision
- Artifacts
 - Final Model Report
 - Evaluation Report



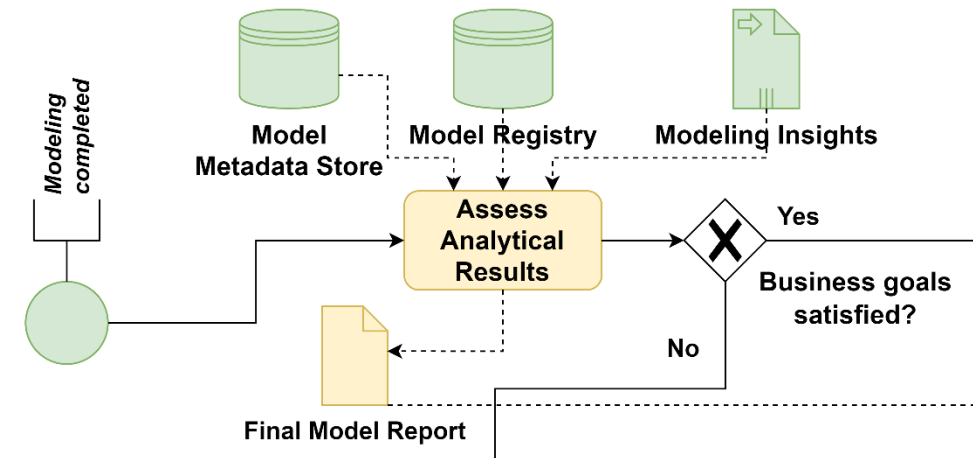
Evaluation – Information Flow Perspective

- Insights from Modeling stage are evaluated with input from domain experts
- Project requirements and process review influence checkpoint decision
- Continuation with Deployment or return to Business Understanding



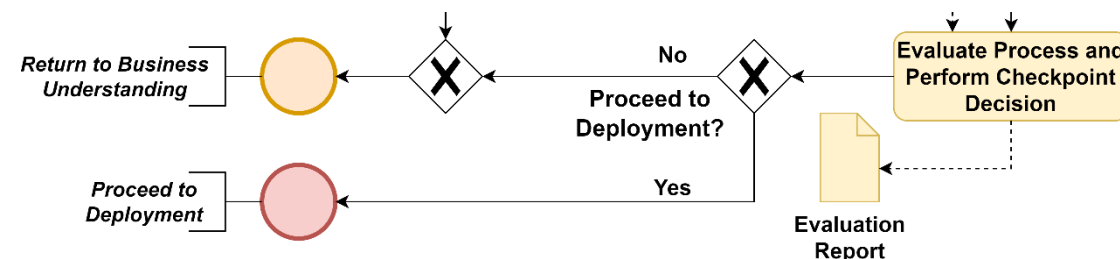
Assess Analytical Results

- Goal: assess the degree to which the model(s) meet the business objectives
- Optional: test model on test applications, if time and budget constraints allow
- Interpret results for application domain and check effect on initial goal
- Final Model Report
 - Summary of technical assessment of considered models
 - Interpretation of achieved DS results → derived insights regarding business objectives and success criteria
 - Ranking of models according to business objectives
 - Potential criteria: feasibility, novelty, generalizability, complexity, limitations, relevance, understandability, transparency, ...



Evaluate Process and Perform Checkpoint Decision

- Goal: review the DS project and determine next steps
- Thorough review of the DS process
 - Necessity of activities and quality of their execution
 - Potential alternatives, failures, and improvements
 - Review of results with respect to business objectives and potential for deployment
- Checkpoint decision on how to proceed with the project based on the review
- Evaluation Report
 - Assessment of project status
 - Business objectives and their fulfillment, including (the necessity of) adjustments or new objectives
 - Checkpoint decision regarding next steps
 - Options: continuation with deployment, further modeling iterations, project abortion, etc.
 - Factors: budget, timeline, fulfillment of objectives, etc.



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- Haertel, C., Pohl, M., Staegemann, D., & Turowski, K. (2022). Project Artifacts for the Data Science Lifecycle: A Comprehensive Overview. 2022 IEEE International Conference on Big Data (Big Data).
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